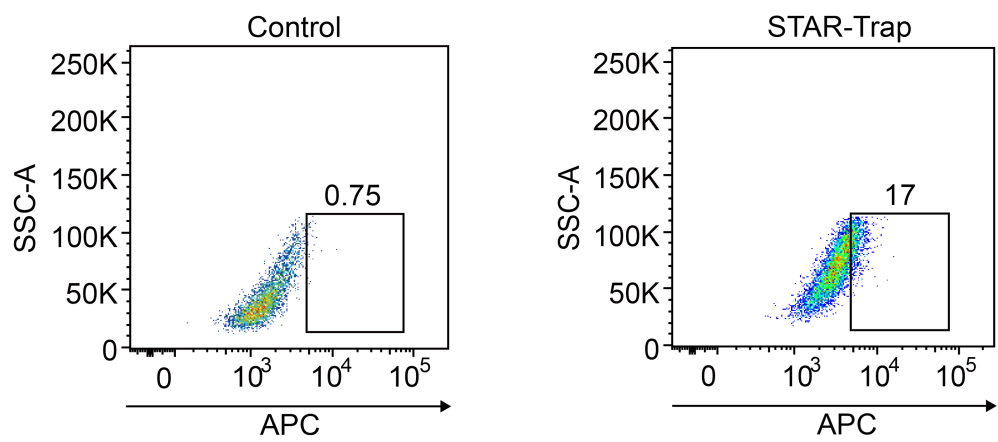


1. **Experiment:** Endocytosis Assay
2. **Time:** 2026.06.29-2026.07.08
3. **Member:** Zheyuan Deng, Ziqian He, Kaili Jin, Xinru Yuan, Zhangdong Liu, Qichang Liu
4. **Materials:** Recombinant human sIL-6R, Recombinant human IL-6, EZ-Link™ Sulfo-NHS-Biotinylation Kit (Thermo Scientific, Cat. No. 21425), PBS buffer, Streptavidin-APC (BD Biosciences, Cat. No. 554067), Pre-chilled PBS, STAR-Traps (untransduced and TOP1-expressing), Complete medium (RPMI-1640 with 10% FBS and 1% penicillin-streptomycin), FACS buffer (PBS with 1% FBS)
5. **Method:**
  - (1) Preparation of STAR-Traps:
    - ① STAR-Traps were cultured in complete medium (RPMI-1640 with 10% FBS and 1% penicillin-streptomycin) at 37 °C in 5% CO<sub>2</sub>.
    - ② STAR-Traps generated and sorted in the preceding experiment were used for the endocytosis assay. Matched untransduced STAR-Traps were used as the cell-type control.
  - (2) Preparation of fluorescently labeled IL-6/sIL-6R complex:
    - ① Recombinant human sIL-6R and IL-6 were mixed at a 1:1 molar ratio and incubated at room temperature for 30 min to form the sIL-6R/IL-6 complex.
    - ② Biotinylation was performed using the EZ-Link™ Sulfo-NHS-Biotinylation Kit (Thermo Scientific, Cat. No. 21425). The complex was buffer-exchanged into PBS and then incubated with a 20× molar excess of Sulfo-NHS-Biotin reagent for 30 min at room temperature in the dark, allowing covalent conjugation of biotin to primary amine groups via stable amide bonds. The sample was centrifuged at 14,000 rpm for 15 min at 4 °C and the supernatant was collected.
    - ③ After the reaction, excess free biotin was removed using Zeba Spin desalting columns to obtain purified biotinylated complex.
    - ④ The biotinylated complex was then incubated with Streptavidin-APC (BD Biosciences, Cat. No. 554067) at a molar ratio of 1:4 for 30 min at 4 °C in the dark to obtain the fluorescently labeled sIL-6R/IL-6 complex.
  - (3) Co-culture of STAR-Traps with the IL-6/sIL-6R $\alpha$  complex:
    - ① STAR-Traps were seeded in 6-well plates at  $1 \times 10^6$  cells/well and cultured for 24 h.
    - ② APC-labeled IL-6/sIL-6R complex was added at a final concentration of 10 nM.
    - ③ Cells were incubated at 37 °C for 2 h. A parallel plate was incubated at 4 °C for 2 h as the internalization-inhibited control.
  - (4) Endocytosis assay and flow cytometric detection of complex-internalizing cells:
    - ① After co-culture, the medium containing unbound complex was removed, and the cells were washed twice with ice-cold PBS.
    - ② Acid wash to remove surface-bound complex: The cells were treated with 0.2 M glycine (pH 2.5) for 5 min at room temperature to remove the complex bound to the cell surface but not internalized, then neutralized with complete medium and washed twice with ice-cold PBS.
    - ③ The cells were detached with a cell scraper, resuspended in FACS buffer (PBS with 1% FBS) at a density of  $1 \times 10^7$  cells/mL, and passed through a 40  $\mu$ m cell strainer to obtain a single-cell suspension.
    - ④ Flow cytometric detection: Gating strategy: single cells  $\rightarrow$  live cells  $\rightarrow$  macrophages  $\rightarrow$  intracellular APC fluorescence. The gating boundary was established using unstained cells, the 4 °C internalization-inhibited control, and the isotype control. APC-positive cells were recorded as cells that had successfully internalized the complex.
    - ⑤ The mean fluorescence intensity (MFI) and the percentage of APC-positive cells were recorded for each group. The internalization efficiency of STAR-Traps was compared with that of wild-type macrophages to evaluate the effect of the STAR6 on the uptake of the IL-6/sIL-6R $\alpha$  complex.

1. Result:



**Figure 1.** Flow cytometric analysis of IL-6/sIL-6R $\alpha$  complex internalization in Control and STAR-Trap. Representative flow-cytometry plots showing uptake of the biotinylated IL-6/sIL-6R $\alpha$  complex labeled with streptavidin-APC by control macrophages (Control) and STAR-Trap after quenching extracellular fluorescence. Numbers indicate the percentages of APC-positive cells within the indicated gates.